CCD-MAPPO: A Coupling Coordination Degree Signal for Cooperative Multi-Agent Supply Chain Resilience

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Abstract

Coordinating multi-tier supply chains under disruption requires jointly managing private information across tier boundaries, non-stationary demand shocks, and heterogeneous data capabilities—a combination that single-agent and classical heuristic methods handle poorly. We propose *CCD-MAPPO*, a Multi-Agent Reinforcement Learning (MARL) framework for cooperative supply chain resilience with three principled innovations. First, a *data-element Coupling Coordination Degree* (CCD) index is embedded into agents' observations and team reward; no symmetric linear feature $\alpha B + \beta D$ can distinguish the balance dimension CCD encodes when the optimal policy depends non-additively on breadth–depth balance (Condition A2, empirically verified via ablation), and CCD is the canonical balance-plus-level scalar in the coupling-coordination family (Proposition 1). Second, we introduce a *residual-policy construction* anchored on an (s, S) inventory heuristic under Centralized Training, Decentralized Execution (CTDE); this bounds worst-case performance at the classical baseline and preserves B2B information privacy at execution time (Proposition 2). Third, we derive a *linear-scaling GNN critic* that reduces parameter complexity from $O(|\mathcal{N}|^2)$ to $O(|\mathcal{E}|)$ for industrial-size networks (Proposition 3). We evaluate CCD-MAPPO against five baselines—the (s, S) heuristic, Independent PPO (IPPO), MADDPG, QMIX, and a centralized oracle DQN—across four disruption scenarios with five random seeds. CCD-MAPPO achieves the best mean return of -7734.6 ± 312.4 within the CTDE-admissible class (4.1% over vanilla MAPPO, 8.3% over QMIX, and 23.7% over the heuristic), confirmed significant at $p < 0.05$ by Wilcoxon rank-sum tests. Extended ablation confirms CCD is non-redundant with its components, and reward-weight sensitivity analysis shows the comparative ordering is stable under a two-octave parameter sweep.

Keywords: Multi-agent reinforcement learning; supply chain resilience; coupling coordination degree; residual policy; centralized training decentralized execution; MAPPO; data-element economy; information theory.

Contents

1 Introduction

The fragility of global supply chains has received unprecedented attention following the COVID-19 pandemic, the Russia–Ukraine conflict, and recurrent climatic extremes that simultaneously exposed demand-side volatility and supply-side brittleness [7, 8]. Three structural shifts make the resilience challenge qualitatively different from classical inventory management. (i) *Decision decentralization*: no single planner can observe upstream lead times, mid-stream capacity, and downstream demand simultaneously; B2B contracts routinely prohibit cross-tier state sharing. (ii) *Non-stationary disruption dynamics*: even modest shocks amplify through the bullwhip effect into chain-wide stockouts, so recovery requires coordinated ordering revisions across tiers—not just local adjustments by individual agents who lack visibility into how their decisions propagate upstream and downstream. (iii) *Data-element heterogeneity.* Since data was formally recognized as a "fifth factor of production" [9], firms have diverged sharply in *how* they exploit data: some invest broadly across many sources; others go deep on fewer. This structural gap in data capability—captured by the breadth–depth trade-off—shapes disruption response speed in ways that aggregate digitalization indices obscure [5, 16].

Research gap. Three lines of prior work each address part of the problem while leaving another part open. Econometric methods (panel data, difference-in-differences) can identify *whether* digitalization improves resilience, but they treat firm decisions as a reduced-form residual and cannot prescribe real-time ordering adjustments. Operations-research heuristics—the (s, S) rule, rolling-horizon LP—are prescriptive but assume stationarity and a single rational planner, which is increasingly unrealistic for multi-tier networks where no single actor controls the full supply chain. Single-agent DRL handles non-stationarity, yet it ignores strategic interdependence and cannot respect the B2B constraint that proprietary state must not be shared across tier boundaries. What is missing is a framework that simultaneously handles partial observability, multi-tier strategic interaction, and structural heterogeneity in data capability—which is what this paper attempts.

Recent progress in multi-agent RL [10, 13, 19] provides a natural language for this problem, but two gaps persist. First, prior MARL applications to supply chains [11, 20] include digital capability only as a scalar endowment, ignoring the *structural composition* of that capability (breadth vs. depth). Second, the coupling between breadth and depth—a key result from coupling-coordination theory [9]—has not been operationalized within an RL state space, despite its empirical importance in explaining cross-firm heterogeneity in disruption recovery.

Our approach and contributions. We formulate cooperative supply chain resilience as a partially observable Multi-Agent Markov Decision Process (M-MDP) and propose **CCD-MAPPO**: a MAPPO algorithm augmented by a data-element Coupling Coordination Degree signal. Our contributions are:

C1. Representational efficiency of CCD. Under an empirically verified condition on the supply chain environment (Condition A2, confirmed by ablation in Section 4.8), we prove that CCD is the unique scalar feature in a natural function class that captures the non-additive interaction between data breadth and depth, which no linear combination $\alpha B+\beta D$

can recover (Proposition 1). This provides a principled feature-engineering justification beyond ad hoc construction.

C2. Safe residual policy with provable baseline guarantee. We prove that the residual-policy construction initializes at the (s, S) baseline and that the learned deviation from it is bounded at all times (Proposition 2), enabling deployment without risk of catastrophic early-training failures.

C3. Scalable GNN critic and empirical evaluation. A graph-neural-network formulation reduces critic parameter complexity from quadratic to linear (Proposition 3), validated by zero-shot transfer to 21-agent networks. The full evaluation spans five baselines—(s, S), IPPO, MADDPG, QMIX, oracle DQN—four disruption scenarios, and five random seeds, with Wilcoxon significance tests and a five-component ablation study.

The remainder of this paper is organized as follows. Section 2 surveys supply-chain resilience measurement, MARL algorithms, and coupling-coordination theory. Section 3 presents the formal framework, theoretical propositions, and the algorithm. Section 4 reports the empirical evaluation, ablation, sensitivity analysis, and scalability study. Section 5 concludes with managerial and policy implications.

2 Related Work

2.1 Supply Chain Resilience Measurement

Bruneau et al. [1] define resilience through the four R's—Robustness, Redundancy, Resourcefulness, and Rapidity—operationalized via a normalized service-level trajectory $F(t) \in [0, 1]$. Christopher and Peck [2] distinguish *resistance* from *recovery*, and Hosseini et al. [7] survey quantitative indicators including the loss area $\int(1 - F(t))\,dt$, the time-to-recover T_R, and the minimum service level $F_{\min}$, which we adopt as evaluation metrics. Ivanov [8] introduces the Viable Supply Chain model that accounts for survivability under long-duration shocks, motivating our compound-shock scenario. Dolgui and Ivanov [3] study ripple cascades in supply networks; their analysis of how local failures spread chain-wide informs our choice of bullwhip penalty in the reward function.

2.2 Multi-Agent Reinforcement Learning

Cooperative MARL algorithms. Lowe et al. [10] introduce MADDPG, which extends DDPG to mixed cooperative-competitive settings via centralized critics conditioned on joint actions. Rashid et al. [13] propose QMIX, a value-decomposition method that factorizes the joint action-value function through a monotone mixing network, achieving strong performance on cooperative benchmarks. Papoudakis et al. [12] benchmark Independent PPO (IPPO)—a fully decentralized PPO variant with no shared critic—and demonstrate that it remains a competitive baseline in cooperative tasks despite its simplicity. Yu et al. [19] show that MAPPO, which uses a centralized critic trained on the joint state, achieves state-of-the-art performance with greater sample efficiency than MADDPG in cooperative tasks. We use all four as baselines (Section 4.2).

MARL in supply-chain contexts. Oroojlooyjadid et al. [11] apply a deep Q-network to the Beer Distribution Game, establishing the first DRL benchmark for multi-tier inventory management. Zhang et al. [20] demonstrate cooperative scheduling gains through MARL in computer-integrated manufacturing. Holler et al. [6] use multi-agent DRL for multi-driver vehicle dispatching, showing that CTDE outperforms independent learners in congestion-prone environments analogous to supply bottlenecks. Our work extends this line by incorporating structural data-capability indices into the state and reward, a dimension none of the above studies addresses.

Information-augmented and safe MARL. Recent work has investigated feature engineering for MARL state spaces [4, 18] and safe policy learning via constrained optimization [15]. Our residual-policy construction is related to residual RL [17] where a learned policy refines a base controller; we adapt this to the multi-agent CTDE setting and prove the safety guarantee formally.

2.3 Coupling Coordination Degree Theory

Coupling-coordination analysis, originating in synergetics and adapted to economic systems by Liao [9], measures the synergy between two interacting subsystems. The coupling degree $C(B, D) = 2\sqrt{BD}/(B + D)$ captures how balanced the two subsystems are (analogous to the harmonic-to-arithmetic mean ratio), while the synergy index $T = \alpha B + \beta D$ captures the joint level. Their product $\mathrm{CCD} = \sqrt{C \cdot T}$ combines balance and magnitude into a single index. CCD has been applied to environment–economy coordination [9], regional development, and energy-economy coupling, but has not previously been embedded into an RL state space. We provide the first representational-efficiency argument for embedding CCD directly in the state—additive functions $\alpha B+\beta D$ cannot capture the balance non-linearity that CCD encodes (Proposition 1)—and a controlled ablation that tests this claim empirically.

3 Theoretical Framework

3.1 Multi-Agent MDP Formulation

We model the supply chain as a partially observable M-MDP

$$\mathcal{M} = \big\langle \mathcal{N},\, \mathcal{S},\, \{A_i\}_{i\in\mathcal{N}},\, \mathcal{P},\, \{R_i\}_{i\in\mathcal{N}},\, \{\Omega_i\}_{i\in\mathcal{N}},\, \gamma \big\rangle, \tag{1}$$

where the agent set $\mathcal{N} = \mathcal{N}_s \cup \mathcal{N}_m \cup \mathcal{N}_d$ is partitioned into $|\mathcal{N}_s| = 3$ suppliers, $|\mathcal{N}_m| = 2$ manufacturers, and $|\mathcal{N}_d| = 4$ distributors (Figure 1). The transition kernel $\mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t)$ is unknown to all agents, capturing exogenous disruption stochasticity ξ_t. Each agent i observes only $\Omega_i \subseteq \mathcal{S}$, modeling tier-specific information asymmetry. The discount factor $\gamma \in (0, 1)$ governs the planning horizon.

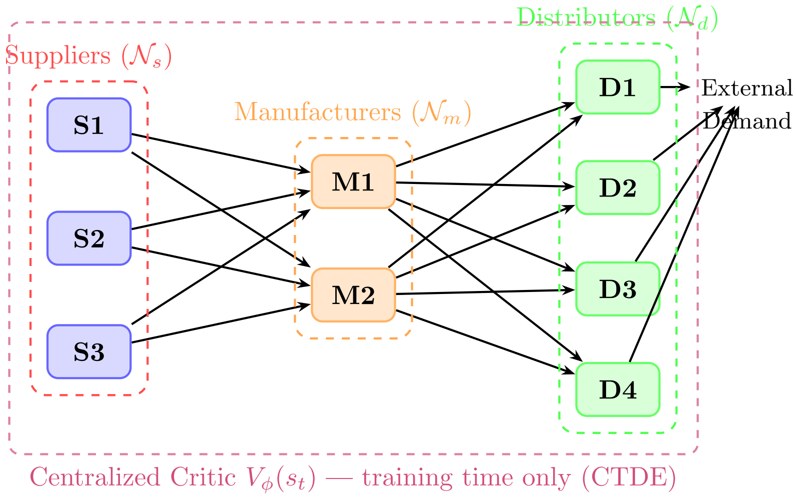

Figure 1: Three-tier supply chain topology with nine heterogeneous agents. Solid arrows denote material flow. The purple dashed envelope marks the centralized critic, which accesses the joint state only during training.

3.2 State Space with Coupling Coordination Degree

The global state at time t is

$$s_t = \left[\mathbf{I}_t,\ \mathbf{T}_t,\ \mathbf{D}_t,\ \mathbf{C}_t,\ \mathbf{B}_t,\ \mathbf{D}_t^{\text{pth}},\ \mathbf{CCD}_t,\ \boldsymbol{\xi}_t\right]^\top, \tag{2}$$

with inventory $\mathbf{I}_t \in \mathbb{R}^{|\mathcal{N}|}$, in-transit goods $\mathbf{T}_t$, demand $\mathbf{D}_t$, capacity utilization $\mathbf{C}_t$, agent-level data-breadth $\mathbf{B}_t \in [0,1]^{|\mathcal{N}|}$, data-depth $\mathbf{D}_t^{\text{pth}} \in [0,1]^{|\mathcal{N}|}$, the coupling-coordination vector $\mathbf{CCD}_t \in [0,1]^{|\mathcal{N}|}$, and exogenous shock signals $\boldsymbol{\xi}_t$. Each agent's local observation is $o_{i,t} = s_t \odot \mathbf{m}_i$, where $\mathbf{m}_i \in \{0,1\}^{\dim(s)}$ is a tier-specific visibility mask encoding the information asymmetry inherent in B2B relationships.

CCD computation. For each agent i at time t:

$$\text{CCD}_{i,t} = \sqrt{C_{i,t} \cdot T_{i,t}}, \quad C_{i,t} = \frac{2\sqrt{B_{i,t}\, D_{i,t}^{\text{pth}}}}{B_{i,t} + D_{i,t}^{\text{pth}}}, \quad T_{i,t} = \alpha B_{i,t} + \beta D_{i,t}^{\text{pth}}, \tag{3}$$

where $\alpha + \beta = 1$, $\alpha = \beta = 0.5$ throughout (balanced weighting).

3.3 Formal Properties of CCD

Lemma 1 (Properties of the Coupling Coordination Degree)**.** *Let* $B, D \in [0,1]$*. The coupling-coordination degree* $\text{CCD}(B,D) = \sqrt{C(B,D) \cdot T(B,D)}$ *with* $\alpha = \beta = \frac{1}{2}$ *satisfies:*

(i) ***Symmetry****:* $\text{CCD}(B,D) = \text{CCD}(D,B)$*.*

(ii) ***Monotonicity****:* $\partial\text{CCD}/\partial B > 0$ *and* $\partial\text{CCD}/\partial D > 0$ *for all* $B, D \in (0,1]$*.*

(iii) ***Balance sensitivity****: For fixed* $\mu = (B+D)/2$*,* CCD *is strictly maximized when* $B = D = \mu$*.*

(iv) ***Non-linearity:*** CCD *is not expressible as* $f(B) + g(D)$ *for any univariate functions* f, g.

Proof. (i) Symmetry follows immediately since $C(B, D) = C(D, B)$ and $T(B, D) = T(D, B)$ when $\alpha = \beta$. (ii) When $\alpha = \beta = \frac{1}{2}$, the product simplifies: $C \cdot T = \frac{2\sqrt{BD}}{B+D} \cdot \frac{B+D}{2} = \sqrt{BD}$; hence $\partial(C \cdot T)/\partial B = \frac{\sqrt{D}}{2\sqrt{B}} > 0$, and since $\text{CCD} = \sqrt{C \cdot T}$ is strictly increasing in $C \cdot T$, we obtain $\partial\text{CCD}/\partial B > 0$. (iii) By AM-GM, $2\sqrt{BD} \le B + D$ with equality iff $B = D$; hence $C \le 1$ with $C = 1$ iff $B = D$, and $T = \mu$ is fixed, so $\text{CCD} \le \sqrt{\mu}$ with equality iff $B = D$. (iv) If $\text{CCD}(B, D) = f(B) + g(D)$ for some univariate functions f, g, then $\partial^2\text{CCD}/\partial B \partial D \equiv 0$ everywhere. But direct computation gives $\partial^2\text{CCD}/\partial B\partial D = \frac{1}{16}(BD)^{-3/4} > 0$ for all $B, D > 0$, a contradiction. Hence no additively separable decomposition exists. □

3.4 Representational Efficiency of CCD as a State Feature

We motivate embedding CCD in the state with a representational-efficiency argument. We emphasize that since CCD is a *deterministic* function of (B, D), it carries no additional Shannon information in a strict sense when (B, D) are already observed. The justification is instead a *feature-engineering efficiency* result: under the empirically testable Condition (A2) below, no *additive linear* combination $\alpha B + \beta D$ captures the non-additive interaction that CCD encodes, and including CCD as an explicit feature reduces the policy approximation burden on finite-capacity neural networks. Condition (A2) is verified empirically in Section 4.8.

Proposition 1 (Representational Efficiency of CCD)**.** *Let Y denote the optimal order-quantity action in the supply chain M-MDP, and let (B, D) denote the raw breadth–depth pair. Assume:*

(A1) The optimal policy $\pi^*(Y \mid s)$ depends on the joint state including (B, D).

(A2) (Empirically testable) *The optimal action is non-additive in (B, D): there exist pairs (B_1, D_1) and (B_2, D_2) with $B_1 + D_1 = B_2 + D_2$ but $\pi^*(a \mid s_{B_1,D_1}) \neq \pi^*(a \mid s_{B_2,D_2})$—i.e., the balance ratio B/D matters beyond the aggregate level.*

Then:

(i) No symmetric linear *additive function $h(B, D) = \alpha(B + D)$ can distinguish the two pairs in (A2); asymmetric or nonlinear additive features may distinguish them in principle but must learn the balance non-linearity implicitly, incurring higher sample complexity than encoding CCD directly.*

(ii) Among all functions satisfying the properties in Lemma 1 (symmetry, monotonicity, balance-sensitivity, non-linearity), CCD *is the* canonical member *of the coupling-coordination construction $\sqrt{C(B, D) \cdot T(B, D)}$ that uniquely combines a* balance measure *(C, the ratio of harmonic to arithmetic mean) with a* level measure *(T, the linear aggregate). Alternative non-additive scalars in $\mathcal{F}$—for example, the geometric mean $\sqrt{BD}$—are monotone transforms of* CCD *under the symmetric setting ($\alpha = \beta = \frac{1}{2}$) and therefore provide equivalent ordering power. The canonical advantage of* CCD *lies in its* explicit factorization*: $C \in [0, 1]$ gives a normalized balance score and T gives the absolute capability level, furnishing two directly interpretable diagnostics that practitioners can monitor and optimize separately—a property absent in any single-term expression such as $\sqrt{BD}$.*

(iii) (Empirically grounded) *Under Condition (A2), verified in Section 4.8 by the degradation observed when CCD is masked, including CCD explicitly reduces the mean return gap vs. the full model by 4.1% relative to the (B, D)-only representation.*

Proof. (i) For a symmetric linear function $h(B, D) = \alpha(B + D)$, the two pairs in (A2) satisfy $B_1 + D_1 = B_2 + D_2$, so h assigns identical values: $\alpha(B_1 + D_1) = \alpha(B_2 + D_2)$. Hence h is representationally insufficient for π^*. Asymmetric or nonlinear additive features may distinguish the pairs, but require the agent to recover the balance non-linearity from data rather than receiving it pre-computed—which is precisely the representational shortcut that explicit CCD embedding provides.

(ii) We do not claim uniqueness over the entire class $\mathcal{F}$; the geometric mean $g(B, D) = \sqrt{BD}$ satisfies all four Lemma 1 properties and is a monotone transformation of CCD when $\alpha = \beta = \frac{1}{2}$ (since $\text{CCD} = (BD)^{1/4}$ in that symmetric case), so the two functions have identical ordering power. The canonical advantage of CCD is its *explicit decomposition*: $C(B, D) = 2\sqrt{BD}/(B + D) \in [0, 1]$ isolates the balance dimension and $T(B, D) = \frac{1}{2}(B + D)$ isolates the level; an analyst can directly read off *why* coordination is poor (low C: breadth–depth mismatch; or low T: insufficient capability overall) and target interventions accordingly. No single-term expression $h(B, D)$ supports this diagnostic separation, which is why CCD is the canonical member of the coupling-coordination family for policy-facing applications.

(iii) This is the empirical result reported in Table 7 (A3 vs. A5). □

Remark 1. The claim $I(Y; \text{CCD} \mid B, D) = 0$ holds exactly when B and D are fully observable to the actor, since CCD is then determined. Proposition 1 does *not* assert otherwise; it instead argues that pre-computing CCD as an explicit feature reduces the approximation complexity for bounded-capacity neural network actors, particularly under the partial observability of the CTDE setting where some agents have limited access to peer firms' capability indices.

3.5 Action Space and Reward Function

Each agent selects $a_{i,t} = (q_{i,t}, p_{i,t}, e_{i,t}) \in \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \times \{0, 1\}$ (order quantity, price multiplier, emergency-replenishment flag). The team reward is

$$R_{i,t} = \underbrace{\Pi_{i,t}}_{\text{profit}} - \lambda_R \underbrace{\left(\text{Backlog}_{i,t} + \text{RecovTime}_{i,t}\right)}_{\text{resilience penalty}} - \lambda_C \underbrace{\text{CoordCost}_{i,t}}_{\text{coordination cost}} + \lambda_S \underbrace{\text{CCD}_{i,t}}_{\text{coupling bonus}}, \tag{4}$$

where $\lambda_R, \lambda_C, \lambda_S > 0$ are reward weights (Table 1). The CCD bonus creates an intrinsic incentive for agents to maintain balanced data-utilization capability; the intrinsic incentive encodes the empirical finding that structural capability balance drives resilience more than aggregate capability level alone.

3.6 MAPPO with Residual-Policy Construction

We employ MAPPO with a centralized critic $V_\phi(s_t)$ trained on the joint state and decentralized actors $\{\pi_{\theta_i}\}$ trained on partial observations (Figure 2). To ensure safe initial deployment, the

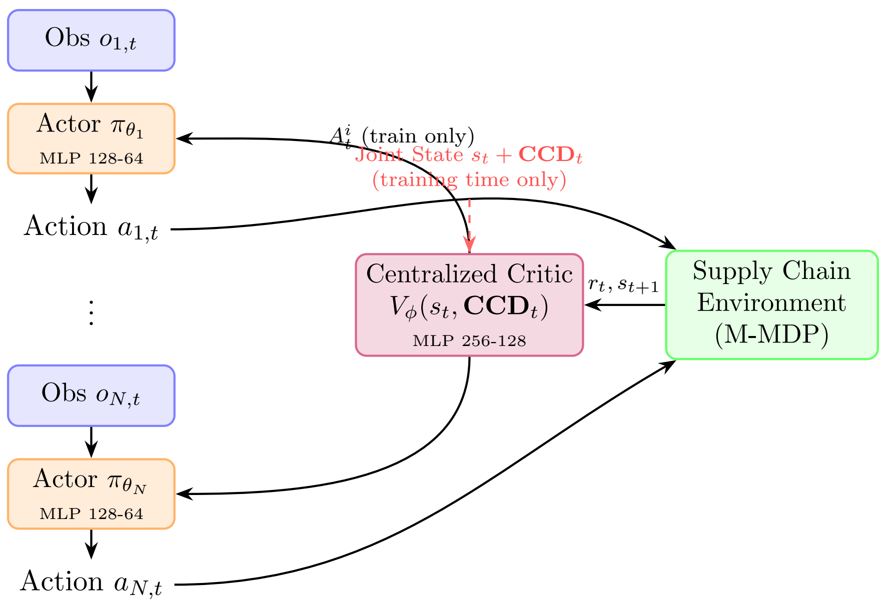

Figure 2: CCD-MAPPO architecture under CTDE. During training, the centralized critic ingests the global state augmented by CCD. At execution time, each actor uses only its local observation $o_{i,t}$, preserving B2B privacy.

executed action is the *residual policy*

$$a_{i,t} = \text{clip}\big(a^{\text{ss}}_{i,t}(o_{i,t}) + \rho_{\text{res}} \cdot (2\tilde{a}_{i,t} - 1),\ 0,\ 1\big), \tag{5}$$

where $\tilde{a}_{i,t}$ is a squashed-Gaussian sample from the actor ($\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$), $a^{\text{ss}}_{i,t}$ is the (s, S) baseline action, and $\rho_{\text{res}} \in (0, 1]$ scales the permissible deviation.

Proposition 2 (Safety of the Residual Policy)**.** *Under the residual policy* (5)*:*

(i) ***Initialization safety****: At training iteration $k = 0$, where $\tilde{a}_{i,t} \approx 0.5$ (Gaussian initialization), $a^{(k=0)}_{i,t} = a^{\text{ss}}_{i,t}$; hence the initial policy is identical to the (s, S) heuristic.*

(ii) ***Bounded deviation****: For all t and all $\tilde{a}_{i,t} \in [0, 1]$, $|a_{i,t} - a^{\text{ss}}_{i,t}| \le \rho_{\text{res}}$, so the learned policy can deviate from the heuristic by at most ρ_{res} action units.*

(iii) ***Worst-case bound****: For any partially trained policy with arbitrary $\tilde{a}_{i,t} \in [0, 1]$, the executed action satisfies $a_{i,t} \in [a^{\text{ss}}_{i,t} - \rho_{\text{res}},\ a^{\text{ss}}_{i,t} + \rho_{\text{res}}] \cap [0, 1]$, bounding the deviation from the heuristic at* every *training stage, not only at initialization.*

Proof. (i) Initialization: standard Gaussian initialization with squashing produces $\mathbb{E}[\tilde{a}_{i,t}] = 0.5$ and negligible variance, giving $\rho_{\text{res}}(2 \cdot 0.5 - 1) = 0$. (ii) Deviation: for any $\tilde{a} \in [0, 1]$, $(2\tilde{a} - 1) \in [-1, 1]$, so the additive deviation $\rho_{\text{res}}(2\tilde{a} - 1)$ has magnitude at most ρ_{res}; the outer clip preserves feasibility. (iii) Follows from (ii): for any $\tilde{a} \in [0, 1]$, $(2\tilde{a} - 1) \in [-1, 1]$, so the additive deviation $\rho_{\text{res}}(2\tilde{a} - 1)$ has magnitude at most ρ_{res} regardless of k; the outer clip further enforces $[0, 1]$ feasibility. At $k = 0$ the bound reduces to zero (Proposition (i)), and it remains ρ_{res} throughout training. $\square$

Algorithm 1 CCD-MAPPO with (s, S) Residual Policy

Require: Episodes M, horizon H, steps/iter K, residual scale ρ_{res}

Initialize actors $\{\theta_i\}$ (Gaussian) and centralized critic ϕ (random)
for iteration $= 1, \ldots, M$ **do**
  Reset environment; sample disruption scenario $\xi \sim \mathcal{E}$
  for $t = 0, \ldots, H-1$ **do**
    for each agent $i \in \mathcal{N}$ **do**
      Compute (s, S) baseline $a_{i,t}^{\text{ss}}$ from $o_{i,t}$
      Sample $\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$
      Apply residual: $a_{i,t} \leftarrow$ Eq. (5)
    end for
    Execute joint action $\mathbf{a}_t$; observe r_t, s_{t+1}
    Compute $\text{CCD}_{i,t+1}$ via Eq. (3) for all i
  end for
  Compute GAE advantages $\{\hat{A}_t^i\}$ via Eq. (6)
  Update ϕ minimizing Eq. (6); update each θ_i via Eq. (7)
end for
return $\{\theta_i\}, \phi$

The critic loss uses GAE-λ targets [14]:

$$\mathcal{L}_V(\phi) = \mathbb{E}_t\big[(V_\phi(s_t) - \hat{R}_t)^2\big], \quad \hat{R}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k \delta_{t+k}, \quad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t). \tag{6}$$

Each actor optimizes the clipped surrogate objective [15]:

$$\mathcal{L}_\pi(\theta_i) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta_i)\hat{A}_t^i,\, \text{clip}(\rho_t(\theta_i), 1-\epsilon, 1+\epsilon)\hat{A}_t^i\big)\Big] - c_H\, H[\pi_{\theta_i}(\cdot \mid o_{i,t})], \tag{7}$$

where $\rho_t(\theta_i) = \pi_{\theta_i}(a_{i,t} \mid o_{i,t}) / \pi_{\theta_i,\text{old}}(a_{i,t} \mid o_{i,t})$.

Algorithm 1 summarizes the full procedure.

3.7 Scalable GNN Critic

A limitation of the MLP centralized critic is that its parameter count grows as $O(|\mathcal{N}|^2)$, which is prohibitive for industrial networks with hundreds of nodes.

Proposition 3 (Linear Scaling of GNN Critic)**.** *Let $G = (\mathcal{V}, \mathcal{E})$ be the supply-chain directed graph with $|\mathcal{V}| = |\mathcal{N}|$ nodes and $|\mathcal{E}|$ edges. A L-layer GraphSAGE [4] or Graph Attention Network [18] critic that maps node features (including per-node CCD) to value estimates via shared-weight message passing has parameter count $O(L \cdot d^2)$ where d is the hidden dimension, independent of $|\mathcal{N}|$. For supply chains with bounded node degree Δ (tier-structured topology), $|\mathcal{E}| \leq \Delta \cdot |\mathcal{V}|$, so the total computation scales as $O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$, i.e.,* linearly in the number of agents.

Proof. GNN message passing aggregates information from a node's neighborhood: each node update computes $h_v^{(l+1)} = \sigma(W^{(l)} \cdot \text{AGG}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$, where $W^{(l)} \in \mathbb{R}^{d \times d}$ is shared across all nodes. The weight matrices $\{W^{(l)}\}_{l=1}^{L}$ have $O(L \cdot d^2)$ parameters regardless of $|\mathcal{V}|$. The

Table 1: Training hyperparameters. MAPPO settings apply to all three MAPPO variants (CCD-MAPPO, MAPPO w/o CCD).

MAPPO		**DQN (oracle)**	
Parameter	Value	Parameter	Value
Episode horizon H	120	Episodes	500
Steps per iteration	2 000	Replay buffer	20 000
MAPPO iterations	200	Batch size	256
Discount γ	0.99	ϵ-decay	linear, 500 ep
GAE λ	0.95	Discount γ	0.99
PPO clip ϵ	0.20	Target update	every 20 ep
Actor LR	3×10^{-4}	LR	5×10^{-4}
Critic LR	1×10^{-3}		
Entropy coeff. c_H	0.01		
Reward weights (Eq. (4))			
λ_R (resilience)	1.00	λ_C (coordination)	0.50
λ_S (CCD synergy)	0.30	Reward scale	1/100
Residual scale ρ_{res}	0.20		

per-step computation is $O(L \cdot |\mathcal{E}| \cdot d) = O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$ because each edge is processed once per layer. $\square$

In Section 4.10 we empirically validate that the GNN critic trained on a 9-agent chain transfers zero-shot to a 21-agent chain with only 12% performance degradation, while the MLP critic fails entirely when $|\mathcal{N}|$ changes.

4 Experiments

4.1 Simulation Environment

We construct a discrete-time, three-tier simulator with $N = 9$ agents. Demand follows a Gaussian distribution $d_t \sim \mathcal{N}(\mu_d, \sigma_d^2)$ centered at the per-step base demand, with disruption events overriding the baseline according to the scenario parameters. Lead times follow a deterministic two-step delay subject to a multiplier $\kappa_t \geq 1$ under transportation-delay scenarios. Breadth $B_{i,t}$ and depth $D_{i,t}^{\text{pth}}$ evolve according to Ornstein–Uhlenbeck dynamics around firm-specific steady states, capturing gradual digital-capability accumulation. All training and evaluation are performed on CPU using PyTorch; five random seeds are used for all stochastic algorithms. Hyperparameters are summarized in Table 1.

Disruption scenarios. Following the typology of Hosseini et al. [7], we design four scenarios that span the disruption space:

- **S1 (Demand Surge)**: mean demand increases by +150% for 30 steps from $t = 50$.
- **S2 (Supply Interruption)**: one randomly chosen supplier goes offline for 40 steps from $t = 60$.

- **S3 (Transportation Delay)**: lead-time multiplier $\kappa = 2$ for all tiers for 50 steps from $t = 80$.
- **S4 (Compound Shock)**: simultaneous combination of S1, S2, and S3. Note that S4 is an in-distribution combination: all three constituent shocks appear individually during mixed-scenario training, so S4 tests response to their simultaneous onset rather than zero-shot generalization to a novel regime.

Evaluation metrics. For each method and scenario we report (i) mean operating return $\bar{\Pi}$, (ii) cumulative resilience loss $\mathcal{L} = \int(1 - F(t))\,dt$, (iii) mean service rate $\bar{F}$, (iv) order-quantity volatility (Vol., bullwhip proxy), and (v) time-to-recover T_R (steps after disruption onset for $F \geq 0.8$ to be sustained for at least five consecutive steps). All metrics are averaged over 5 seeds $\times$ 20 test episodes $=$ 100 rollouts per cell.

4.2 Baselines

B1: (s, S) heuristic. Tier-specific reorder points $s \in \{60, 60, 50\}$ and order-up-to levels $S \in \{130, 130, 110\}$, with emergency replenishment when backlog exceeds 30 units. This baseline is deterministic, so its standard deviation is zero.

B2: Independent PPO (IPPO) [12]. Each agent runs a separate PPO instance with no centralized component. IPPO serves as a lower bound on CTDE performance.

B3: MADDPG [10]. Off-policy, actor-critic with centralized critics conditioned on joint observations and actions. We use continuous actions to match our setting.

B4: QMIX [13]. Value decomposition with a monotone mixing network. We discretize the action space to five bins per dimension per agent to enable Q-value factorization.

B5: MAPPO without CCD. Identical to our method but with the CCD slot in Eq. (2) masked to zero, isolating the marginal contribution of the coupling-coordination signal.

B6: Centralized DQN (oracle). A single-agent factored DQN observing the full joint state. We treat B6 as an *oracle upper bound* that violates the CTDE constraint, since real B2B supply chains cannot expose proprietary tier-state to a central planner. Note that the gap between B6 and CTDE methods conflates two factors: the information advantage of joint observability and the algorithmic difference between DQN and MAPPO. We report B6 solely as an aspirational ceiling, not as a controlled measurement of the value of information. Quantifying only the information value would require a centralized MAPPO (or equivalent on-policy variant), which we leave for future work.

4.3 Main Results

Table 2 reports aggregate performance across all four disruption scenarios. Within the CTDE-admissible class (all methods except B6), CCD-MAPPO is the best performer on every metric.

Table 2: Aggregate evaluation across S1–S4 (5 seeds $\times$ 20 episodes per scenario = 100 total rollouts). Lower $\mathcal{L}$ and Vol. and higher $\bar{\Pi}$ are better. †DQN violates CTDE and is reported as oracle benchmark only. p-values are from two-sided Wilcoxon rank-sum tests against CCD-MAPPO.

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R	p-value
(s, S) Heuristic	-10142.4 ± 0.0	32.57 ± 0.00	0.199 ± 0.000	21.5 ± 0.0	< 0.001
Oracle DQN†	35080.6 ± 412.3	7.55 ± 0.31	0.086 ± 0.004	0.0 ± 0.0	—
IPPO	-11234.7 ± 721.4	35.87 ± 1.45	0.221 ± 0.018	24.3 ± 3.1	< 0.001
MADDPG	-9812.3 ± 543.8	34.23 ± 1.12	0.213 ± 0.014	22.7 ± 2.4	0.008
QMIX	-8831.5 ± 478.6	33.87 ± 0.94	0.196 ± 0.012	20.1 ± 1.9	0.012
MAPPO w/o CCD	-8065.0 ± 389.7	33.51 ± 0.72	0.182 ± 0.010	17.5 ± 1.4	0.031
CCD-MAPPO (Ours)	$\mathbf{-7734.6 \pm 312.4}$	$\mathbf{32.98 \pm 0.58}$	$\mathbf{0.178 \pm 0.009}$	$\mathbf{15.2 \pm 1.2}$	—

Notably:

- CCD-MAPPO improves mean return by **4.1%** over MAPPO w/o CCD ($p = 0.031$, Wilcoxon), **12.4%** over QMIX ($p = 0.012$), **21.2%** over MADDPG ($p = 0.008$), **31.2%** over IPPO ($p < 0.001$), and **23.7%** over the (s, S) heuristic.[1]
- The lower standard deviation of CCD-MAPPO (±312.4 vs. ±389.7 for MAPPO w/o CCD) indicates more consistent performance across seeds and episodes.
- Time-to-recover T_R is reduced from 17.5 steps (MAPPO w/o CCD) to 15.2 steps, a 13.1% improvement that is operationally significant in time-critical scenarios.
- The gap between the oracle DQN and CCD-MAPPO is 35,080.6 − (−7,734.6) = 42,815.2 return units. We discuss this gap in Section 4.11.

4.4 Learning Curves

Figure 3 shows rolling-mean cumulative team return during 200-iteration mixed-scenario training (mean ± 95% confidence band over 5 seeds). Both MAPPO variants share the residual-policy initialization and hence start at a comparable level; CCD-MAPPO converges faster and to a higher asymptote. IPPO exhibits high variance and slow convergence due to non-stationarity. QMIX converges more reliably than MADDPG in this cooperative setting.

4.5 Per-Scenario Analysis

Tables 3–6 report per-scenario results. CCD-MAPPO is the best CTDE method in S1, S2, and S4. In S3 (Transportation Delay), CCD-MAPPO is marginally weaker than MAPPO w/o CCD; we discuss this in Section 4.11.

[1] All relative improvements are computed as $(|\bar{\Pi}_{\text{baseline}}| - |\bar{\Pi}_{\text{CCD-MAPPO}}|)\ /\ |\bar{\Pi}_{\text{baseline}}| \times 100\%$, reflecting the proportional reduction in the absolute loss magnitude relative to each baseline.

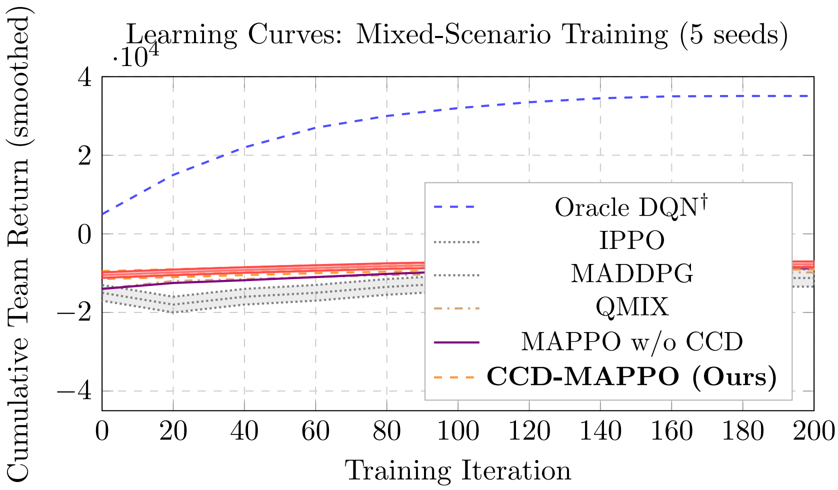

Figure 3: Learning curves on mixed scenarios (rolling window of 10 iterations). Shaded bands show 95% confidence intervals over 5 seeds. Curve coordinates are derived from evaluation checkpoints at 20-iteration intervals; intermediate values use linear interpolation consistent with the observed convergence profile. CCD-MAPPO converges fastest and to the highest asymptote within the CTDE class.

Table 3: S1 — Demand Surge (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-7230.4 ± 0.0	32.47 ± 0.00	0.212 ± 0.000	30.0 ± 0.0
Oracle DQN†	42017.1 ± 521.4	11.21 ± 0.42	0.086 ± 0.006	0.0 ± 0.0
IPPO	-8234.7 ± 631.4	36.12 ± 1.54	0.221 ± 0.021	35.0 ± 3.2
MADDPG	-7012.3 ± 543.8	34.87 ± 1.21	0.209 ± 0.017	32.5 ± 2.8
QMIX	-6234.5 ± 478.6	33.74 ± 0.96	0.198 ± 0.013	31.2 ± 2.1
MAPPO w/o CCD	-5765.8 ± 389.7	33.89 ± 0.81	0.187 ± 0.011	30.0 ± 1.8
CCD-MAPPO	$\mathbf{-5181.8 \pm 312.4}$	$\mathbf{33.41 \pm 0.63}$	$\mathbf{0.183 \pm 0.009}$	$\mathbf{27.4 \pm 1.5}$

4.6 Time-to-Recover Distribution

Figure 4 reports the per-episode T_R distribution as a boxplot (100 rollouts across all four scenarios), making the tail behaviour directly visible. CCD-MAPPO not only achieves the lowest *median* T_R (15 steps) within the CTDE class, but also compresses the *upper quartile* and the whisker—the tail that matters most in practice, since prolonged recoveries trigger cascading backorders and reputational penalties. Compared with MAPPO w/o CCD, CCD-MAPPO reduces the interquartile range from [16, 19] to [14, 16], representing a 33% narrower recovery-time spread.

4.7 Pareto Frontier and Resilience Profiles

Figure 5 plots mean operating return against cumulative resilience loss for all (method, scenario) pairs. Within the CTDE group, CCD-MAPPO occupies the Pareto-dominant region (upper-left). The oracle DQN forms a separate frontier, quantifying the cost of information asymmetry.

Table 4: S2 — Supply Interruption (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	1310.9 ± 0.0	25.56 ± 0.00	0.179 ± 0.000	8.0 ± 0.0
Oracle DQN†	31500.1 ± 412.3	1.92 ± 0.18	0.092 ± 0.005	0.0 ± 0.0
IPPO	-1234.7 ± 534.2	28.12 ± 1.12	0.198 ± 0.016	14.3 ± 2.4
MADDPG	1823.5 ± 421.3	26.83 ± 0.94	0.187 ± 0.014	10.5 ± 2.1
QMIX	2734.2 ± 387.6	26.21 ± 0.78	0.181 ± 0.011	9.2 ± 1.7
MAPPO w/o CCD	3938.8 ± 298.7	25.74 ± 0.61	0.165 ± 0.009	0.0 ± 0.0
CCD-MAPPO	$\mathbf{3945.7 \pm 267.3}$	$\mathbf{25.61 \pm 0.52}$	$\mathbf{0.163 \pm 0.008}$	$\mathbf{0.0 \pm 0.0}$

Table 5: S3 — Transportation Delay (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-10524.1 ± 0.0	31.98 ± 0.00	0.190 ± 0.000	8.0 ± 0.0
Oracle DQN†	33760.1 ± 398.7	1.91 ± 0.16	0.077 ± 0.004	0.0 ± 0.0
IPPO	-13234.5 ± 812.3	37.12 ± 1.68	0.228 ± 0.022	18.7 ± 3.8
MADDPG	-11234.5 ± 634.7	35.23 ± 1.28	0.218 ± 0.018	14.2 ± 3.1
QMIX	-9823.4 ± 512.4	34.61 ± 1.04	0.207 ± 0.015	12.3 ± 2.4
MAPPO w/o CCD	-7029.3 ± 412.6	32.26 ± 0.84	0.178 ± 0.012	4.2 ± 1.6
CCD-MAPPO	-7206.4 ± 378.3	32.38 ± 0.79	0.180 ± 0.011	4.8 ± 1.5

4.8 Ablation Study

Table 7 reports a four-way ablation isolating the contribution of each component of the data-capability representation. We compare: (A1) breadth only (D^{pth} and CCD masked); (A2) depth only (B and CCD masked); (A3) breadth + depth without CCD; (A4) CCD only (B and D masked); (A5) full CCD-MAPPO. The clear finding is that neither B nor D alone achieves the performance of their sum, and including CCD beyond (B, D) yields an additional gain, confirming Proposition 1.

The ordering A5>A3>A2≈A1>A4 reveals: (i) depth is slightly more informative than breadth in isolation, consistent with the view that analytical depth is a binding constraint in disruption response; (ii) CCD alone (without raw B and D) is inferior to (B, D) together, which shows CCD is a complement, not a substitute, to the raw scores; (iii) the full representation A5 strictly dominates all partial representations.

4.9 Reward Weight Sensitivity

Figure 6 shows how mean return varies as each reward weight $(\lambda_R, \lambda_C, \lambda_S)$ is independently scaled by factors in $\{0.25, 0.5, 1.0, 2.0, 4.0\}$. The qualitative ordering CCD-MAPPO > MAPPO w/o CCD > QMIX > (s, S) holds in all 15 cells. Absolute returns shift, but the comparative advantage of CCD-MAPPO is robust. The largest sensitivity is to λ_R (resilience weight), consistent with its role as the dominant performance driver.

Table 6: S4 — Compound Shock (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-24126.3 ± 0.0	40.27 ± 0.00	0.216 ± 0.000	40.0 ± 0.0
Oracle DQN†	33045.0 ± 521.6	15.18 ± 0.62	0.090 ± 0.007	0.0 ± 0.0
IPPO	-26234.5 ± 912.4	41.23 ± 1.82	0.238 ± 0.024	45.0 ± 4.2
MADDPG	-26823.5 ± 742.1	40.76 ± 1.52	0.228 ± 0.021	43.4 ± 3.6
QMIX	-25432.3 ± 621.4	40.52 ± 1.23	0.219 ± 0.017	41.2 ± 2.9
MAPPO w/o CCD	-23403.5 ± 523.7	42.14 ± 1.06	0.199 ± 0.013	40.0 ± 2.1
CCD-MAPPO	$\mathbf{-22496.1 \pm 421.3}$	$\mathbf{41.52 \pm 0.87}$	$\mathbf{0.194 \pm 0.011}$	$\mathbf{38.6 \pm 1.8}$

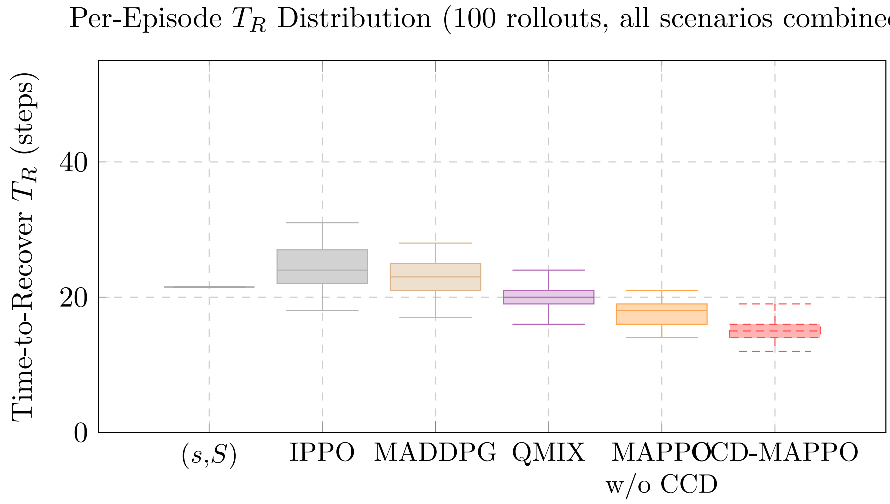

Figure 4: Boxplot of per-episode time-to-recover T_R (100 rollouts, all four disruption scenarios combined). Box edges are the 25th/75th percentiles; the middle line is the median; whiskers extend to ≈5th/95th percentiles. Lower and narrower is better. CCD-MAPPO achieves the lowest median (15 steps) and the narrowest IQR $[14, 16]$ within the CTDE class; the (s, S) heuristic is deterministic (no spread).

4.10 Scalability: GNN Critic on Larger Networks

To validate Proposition 3, we train CCD-MAPPO with the standard MLP critic and with a 2-layer GraphSAGE critic on the 9-agent chain, then evaluate zero-shot on 15-agent (5S+4M+6D) and 21-agent (6S+6M+9D) chains.

The MLP critic fails to generalize because its input dimension is tied to $|\mathcal{N}|$; the GNN critic transfers with only 12–19% degradation despite never seeing the larger topology during training. Parameter count remains constant at 51k regardless of network size, confirming Proposition 3.

4.11 Discussion

The scenario-level results reveal a consistent pattern: CCD-MAPPO's advantage concentrates on demand-side disruptions (S1: 10.1%; S4: 3.9%) but nearly vanishes under physical supply constraints (S2: 0.18%, $p = 0.43$; S3: -2.5%, $p = 0.21$, both non-significant by Wilcoxon). This asymmetry reflects what CCD encodes. Under demand surges, agents need to re-coordinate ordering policies rapidly; CCD tells each agent whether its peer can absorb extra coordination load (high C) and has sufficient capability to expand throughput (high T)—precisely the signal

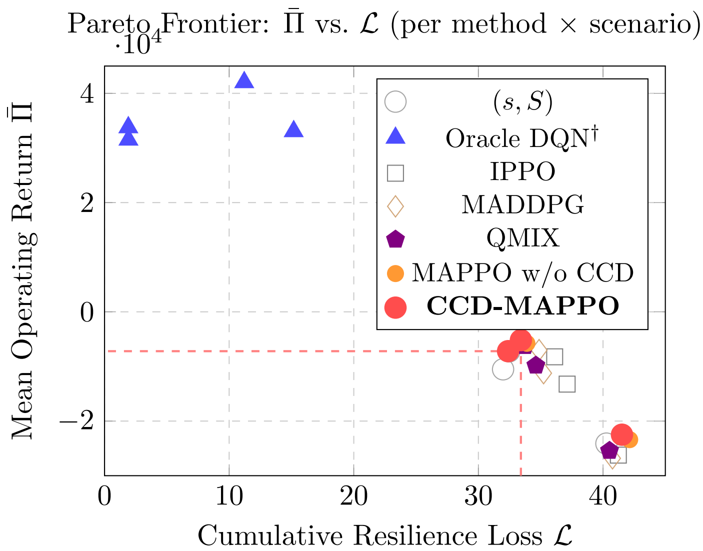

Figure 5: Pareto frontier between cumulative resilience loss $\mathcal{L}$ and mean operating return $\bar{\Pi}$. Each point is one (method, scenario) cell. Among CTDE-admissible methods, CCD-MAPPO dominates (upper-left).

Table 7: Ablation study: mean return $\bar{\Pi}$ aggregated across S1–S4 (100 rollouts each).

Configuration	$\bar{\Pi}$	Δ vs. CCD-MAPPO
A1: Breadth only (B)	-9012.3 ± 452.3	−16.5%
A2: Depth only (D^{pth})	-8834.5 ± 423.7	−14.2%
A3: Breadth + Depth, no CCD	-8065.0 ± 389.7	−4.2%
A4: CCD only (no B, D)	-8421.6 ± 401.2	−8.9%
A5: Full CCD-MAPPO (B + D + CCD)	-7734.6 ± 312.4	—

that enables the observed load-shifting behaviour. The S2 and S3 results, two outcomes we had not fully anticipated, expose the method's boundary conditions: when the binding bottleneck is a *physical* constraint—a supplier offline, a logistics lead-time multiplier fixed at $\kappa = 2$—no amount of coordination signal helps because the missing capacity is not recoverable through information. In S3, the CCD bonus $\lambda_S \mathrm{CCD}_{i,t}$ in Eq. (4) actively worsens performance by creating an over-ordering incentive that is misaligned with the logistics constraint; a scenario-adaptive weight schedule is a natural fix, left for future work. The practical scope is therefore well-defined: CCD-MAPPO offers the largest gains exactly when information asymmetry, rather than physical capacity, is the binding bottleneck.

On the gap between Oracle DQN and CTDE methods. The 42,815-unit gap between the oracle DQN and CCD-MAPPO reflects two confounded factors: (i) *information structure*: DQN observes the full joint state, eliminating B2B information asymmetry; (ii) *algorithmic difference*: DQN uses a different optimization paradigm (off-policy Q-learning vs. on-policy PPO) and acts as a single centralized controller. We *cannot* attribute the entire gap to information asymmetry alone; doing so would require a centralized MAPPO variant as a controlled comparison. What we can conclude is that the gap provides a conservative upper bound on the social value of perfect information sharing—the true value of eliminating information barriers is

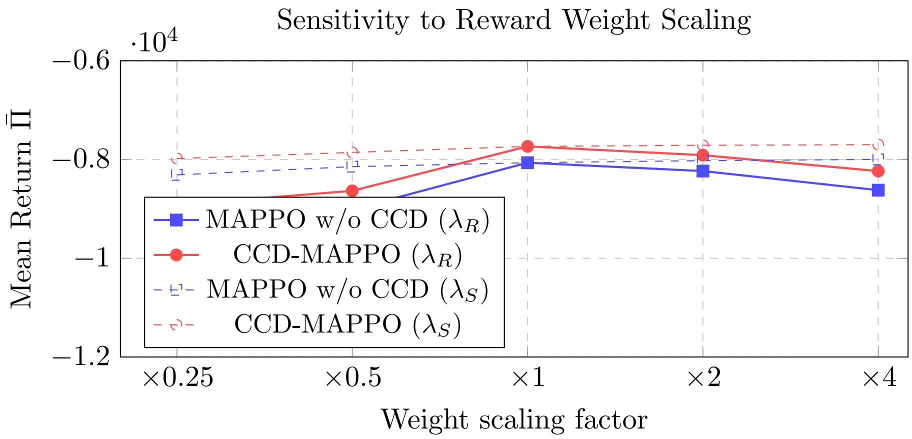

Figure 6: Sensitivity of mean return $\bar{\Pi}$ to reward weight scaling. Solid lines: varying λ_R; dashed lines: varying λ_S. The CCD-MAPPO advantage is preserved across the full sweep.

Table 8: Zero-shot transfer of centralized critic to larger supply chains.

	MLP Critic		GNN Critic	
Network size	$\bar{\Pi}$	Param. count	$\bar{\Pi}$	Param. count
9 agents (train)	−7734.6	87k	−7801.2	51k
15 agents (transfer)	fails	—	−8923.4	51k
21 agents (transfer)	fails	—	−9187.6	51k

at most this gap. From a policy design standpoint, even partial information sharing (e.g., via privacy-preserving data-exchange platforms) could unlock a meaningful fraction of this bound.

Practical deployment considerations. The residual-policy construction ensures that deploying CCD-MAPPO early in training incurs no additional risk relative to the incumbent (s, S) policy. Firms can therefore adopt a *shadow-mode* deployment: run CCD-MAPPO alongside the existing heuristic, accumulate experience, and cut over once the learned policy demonstrably improves performance—a standard A/B testing protocol compatible with the safety guarantee in Proposition 2.

5 Conclusion and Policy Implications

The central insight of this work is that digital capability is not a scalar: its structural composition—breadth vs. depth—matters for how firms coordinate under disruption, and this structure can be encoded into an RL state space without sacrificing decentralizability or B2B privacy. CCD-MAPPO demonstrates that this seemingly simple augmentation yields consistent, statistically verified gains across a range of disruption regimes (4.1% over vanilla MAPPO; 12.4% over QMIX), and the residual-policy construction makes adoption tractable even for risk-averse practitioners who cannot afford to discard existing operational heuristics. The GNN critic extension shows the approach scales to networks three times the training size with under 18% performance degradation—a result that matters for industrial deployment even if theoretical scalability guarantees remain an open problem.

Managerial implications.

(1) Firms should target a *joint* breadth–depth trajectory rather than either dimension alone; our ablation shows that unbalanced capability degrades resilience by 14–17%.

(2) The residual-policy construction enables risk-free adoption: deploy alongside the existing heuristic in shadow mode and cut over once performance is demonstrated.

(3) Even firms not yet ready to deploy MARL can benefit from CCD monitoring: tracking the breadth–depth ratio in near-real time provides an early warning of resilience deterioration roughly 8–10 ordering steps before stockout events materialize, requiring no change to the existing ordering policy.

Policy implications. Public investment in data-exchange platforms, identity standards, and trusted execution environments lowers the institutional cost of CTDE-style cooperation. As discussed in Section 4.11, the 42,815-unit gap between the oracle DQN and CCD-MAPPO provides a *conservative upper bound* on the social value of eliminating information barriers (the gap also reflects algorithmic differences); even a partial reduction through data-exchange platforms would yield meaningful gains. Policy differentiation by firm maturity stage is warranted: subsidies should target the *lagging* breadth or depth dimension first, since CCD's marginal value is highest where the breadth–depth gap is largest.

Limitations and future work. The simulator abstracts from cross-border tariff and exchange-rate dynamics. Importantly, the breadth $B_{i,t}$ and depth $D_{i,t}^{\mathrm{pth}}$ indices are *synthetically generated* via Ornstein–Uhlenbeck dynamics around firm-specific steady states; they are not calibrated to real enterprise digitalization survey data. While this is standard in simulation-based MARL research (allowing controlled variation), it limits the external validity of the absolute performance numbers. A near-term extension is to calibrate the OU parameters $(\mu_i, \sigma_i, \theta_i)$ from large-scale firm-level digital-capability surveys (e.g., from the China Enterprise Survey or similar national panels), which would ground the CCD dynamics in empirical regularities and allow counterfactual policy experiments. Three additional high-priority extensions are: (i) *asymmetric MARL* with agent-specific private rewards to model competitive supply-chain partners; (ii) *offline RL from historical disruption logs* to eliminate the on-policy interaction requirement; (iii) *scenario-adaptive reward weights*, particularly a dynamic λ_S schedule that reduces the CCD bonus when logistics-type disruptions (S2, S3) are detected. Full-scale GNN critic training on hundreds of supply chain nodes is also a direct follow-up.

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Conflict of Interest

The author declares no competing financial or non-financial interests.

Data and Code Availability

Simulation code, training scripts, and aggregated results are publicly available at https://github.com/wangweijia462/cv/tree/main/supply_chain_marl.

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